Work Division Plan

UE24CS352A Machine Learning Mini-Project — Group 22 — Problem 22: Generating Music with Machine Learning

Role	Member	SRN
Member A	Nagashree R Patil	PES1UG24CS289
Member B	Nallamalli Kanaka Mani Sai Akhil	PES1UG24CS290

1. Principle

- Both members run the full notebook once so each understands the whole pipeline (graded in Q&A).
- Each member **owns** a part below, commits it to the shared GitHub repo under their own account, and presents it in the review.
- The split mirrors the reference paper, where each member owned specific models plus a write-up/survey task.

2. Ownership split

Area	Member A — Nagashree (Data, Baseline, Evaluation, Documentation)	Member B — Akhil (Models, Generation, Survey)
Code	• Data download, MIDI parsing, event representation, vocabulary, train/val/test windows • Markov (bigram) baseline • Evaluation: loss, perplexity, top-1/top-5, results table, loss-curve plots	• LSTM and GRU model classes and training loop (early stopping, clipping) • Hyper-parameter experiments (hidden size, dropout, learning rate) • Generation (temperature / top-k), MIDI export, audio playback
Experiments	Unknown-token rate, vocabulary size and sequence-length sanity checks; compare models in one table	Generate clips from each model; try 2-3 temperature values and pick the best-sounding setting
Survey	Analyse survey responses, add the numbers to the results table	Create the Google Form (4 clips, "human or computer?"), collect at least 15 responses
Repo	Create the private repo, add faculty/TAs as collaborators, write README, requirements.txt, .gitignore	Add generated/ clips, tag the final commit, verify the notebook runs from a fresh Colab session
Write-up (PDF)	Sections 1-3 (problem, dataset, approach) and results table; final PDF export	Section 4 (implementation) and Section 5 (conclusions / future work); proofread
Slides	Problem, dataset and preprocessing, baseline, results slides	Neural models, generation demo, conclusions and future-work slides
Live demo	Walk through data pipeline and results	Run live generation and play the clips

3. Suggested order of work

Step	Task	Who
1	Upload notebook to Colab (T4 GPU) and run end to end; fix any errors	Both (A drives, B reviews)
2	Save results.csv, loss_curves.png, generated/*.mid; create repo and push	A
3	Create the listening survey with the generated clips and share it with classmates	B
4	Hyper-parameter / temperature experiments; re-run if results improve	B
5	Fill the results table and write the PDF write-up	A (sections 1-3), B (sections 4-5)
6	Build slides; rehearse the demo and cross-ask Q&A questions	Both

Step	Task	Who
7	Final check against the submission checklist and submit before the deadline	Both

4. Questions each member should be ready to answer

- **Both:** Why per-song splitting? Why does the vocabulary cap create an unknown token? What is perplexity? What does temperature do in sampling?
- **Nagashree:** How is a MIDI file turned into events? Why 0.125 s quantisation and at most 4 notes per chord? How does the Markov baseline work and why is it weaker?
- **Akhil:** How does an LSTM differ from a GRU? What do dropout, gradient clipping and early stopping prevent? How is a new piece generated from a seed?

5. Final submission checklist (deadline: Sat 10 Oct 2026, 11:59 PM)

- Private GitHub repo shared with faculty (and TAs); README with setup and run steps; commits from both members
- PDF write-up, kept to the length the faculty specify (guidelines say both "one-page" and "two-page": confirm with faculty)
- Slide deck and a working live demo for the review session
- Generated .mid files, results table and plots included in the repo